

Extraction of Silicon Strip-Waveguide Group Index Using Four Unbalanced Mach–Zehnder Interferometers

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edX public username: HuynhWSiPho (inferred from layout labels)

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Abstract

This proposal describes four passive silicon Mach–Zehnder interferometers (MZIs) contained in `EBeam_HuynhWSiPho.gds`. The objective is to extract the quasi-TE waveguide group index from the free spectral range (FSR) and compare it with a simulated waveguide model. Each circuit uses nominally 500 nm-wide strip waveguides, a Y-branch input splitter, a broadband directional-coupler combiner, and three TE grating couplers. The common short arm is 31.282 μm ; the four arm imbalances are 68.717, 118.719, 168.717, and 58.717 μm . With the course-specified 220 nm silicon thickness and an approximate oxide-clad effective-index model, $n_g(1550\text{nm}) \approx 3.8935$. The corresponding FSRs range from 3.66 to 10.51 nm, providing several fringes within a 50 nm scan. A compact polynomial model, normalized circuit spectra, and a numerical check of the group-index extraction are presented. The simulations are reduced-order estimates, not Lumerical outputs or experimental measurements. Routing markers and short bend-adjacent segments require review before fabrication.

1 Introduction and design objectives

The group index sets the delay per unit length of a guided optical signal and is a useful quantity for designing delay lines, filters, interferometers, and resonators. Whereas the effective index determines accumulated phase, the group index describes how that phase changes with optical frequency. An unbalanced interferometer converts this frequency-dependent phase into measurable transmission fringes. Its FSR therefore provides a practical way to characterize the waveguide without directly measuring propagation time.

The objectives are to (i) measure the quasi-TE group index of a nominal 500 nm $\times$ 220 nm silicon strip waveguide, (ii) verify the inverse dependence of FSR on arm imbalance, and (iii) compare the extracted group index and its wavelength dependence with waveguide simulations. The selected devices vary path length while keeping nominal width, polarization, and component types fixed. This separates the path-length trend from an intentional width sweep.

The supplied file contains four designs, rather than the recommended five to ten. Four imbalances are sufficient to test the principal FSR trend. If an updated layout is permitted, a

duplicate MZI and a grating-to-grating reference waveguide would provide useful fabrication-repeatability and normalization data. They are proposed additions and are not present in the uploaded GDS.

2 Layout-derived circuit design

2.1 Evidence and assumptions

The physical top cell is `EBeam_HuynhWSiPho.v1`. The additional `CONTEXT.INFO` top cell contains `KLayout` context references and is excluded from the circuit inventory. The GDS user unit is 1 μm and its database precision is 1 nm. The layout contains MZI1–MZI4 in a 605 $\mu\text{m} \times$ 410 μm floorplan. Its silicon geometry and routing-marker overlays are shown in Fig. 1.

The waveguide labels explicitly specify `Strip TE 1550 nm`, `w=500 nm`, a 500 nm width, and a nominal bend-radius parameter of 5 μm . Silicon thickness is not encoded by a planar mask; $h = 220\text{nm}$ is taken from the course requirements. An oxide upper and lower cladding is assumed for the numerical model, consistent with the oxide grating-coupler cell name and the published OpenEBL process description [3]. Actual thicknesses, oxide coverage, and material dispersion must be confirmed for the assigned fabrication run. No simulation project, mode export, fabrication data, or measured spectrum is included in the GDS.

2.2 Components and optical connections

Each MZI contains one `ebeam_y_1550` Y-branch, one `ebeam_bdc_te1550` broadband directional coupler, and three `ebeam_gc_te1550` TE grating couplers. The input grating feeds the Y-branch. Its two outputs connect through unequal waveguide arms to the two input ports of the directional coupler, which is rotated by 90°. Two further waveguides connect the directional-coupler outputs to the output gratings. The circuit is therefore a Y-branch/broadband-coupler MZI, not a circuit with two Y-branches.

The directional-coupler label identifies a nominal 50/50 TE splitter at 1550 nm and cites the asymmetric-waveguide phase-control design of Lu *et al.* [5]. The grating cell references

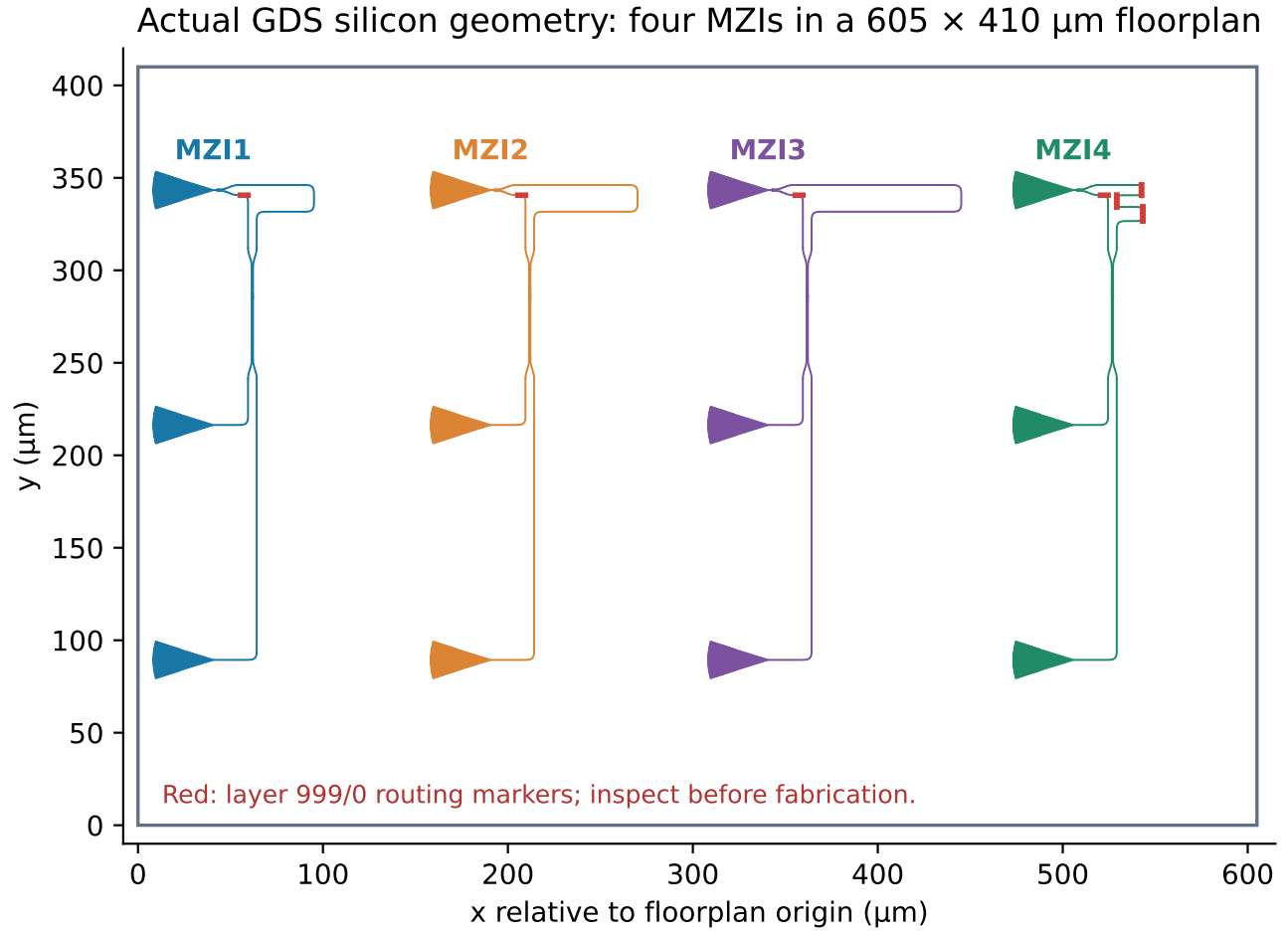


Figure 1: Silicon polygons extracted directly from the supplied GDS. Devices appear from left to right as MZI1, MZI2, MZI3, and MZI4. The frame is the actual $605 \mu\text{m} \times 410 \mu\text{m}$ floorplan. Red overlays indicate paths on layer 999/0, which coincide with short routing segments and require review; they are not additional optical waveguides.

Wang *et al.* [6]; these references identify the component design lineage rather than establish the loss or bandwidth of the specific fabricated devices. No measured component S-parameters were supplied.

For every device, the local input grating port is at $(39.969, 267.500) \mu\text{m}$. The output grating ports are at $(39.969, 140.500) \mu\text{m}$ and $(39.969, 13.500) \mu\text{m}$, giving $127 \mu\text{m}$ pitch. Figure 2 shows two representative circuits. Layer-10 input labels have the form `opt_in_TE_1550_device.HuynhWSiPho_MZI1`, with the device number changed for each cell. The report uses HuynhWSiPho as the provisional edX public username because it appears in all four measurement labels and the filename.

2.3 Arm lengths and parameter variations

The relevant lengths are the two routes between the Y-branch output ports and the directional-coupler input ports. They were read from the embedded `Spice_param:wg_length` labels and corroborated by the `Length` labels, rather than estimated from bounding boxes or sharp-corner waypoint distances. They

are PCell-reported route lengths, conditional on the generated geometry being accepted; they are not independently measured fabricated lengths.

The common short-arm cell Waveguide is $31.282 \mu\text{m}$ long. Long-arm cells are Waveguide\$4, Waveguide\$5, Waveguide\$6, and Waveguide\$8 for MZI1–MZI4. The output access routes are 42.378 and $174.078 \mu\text{m}$ long. Their difference is not included in ΔL : they occur after recombination and cannot create the principal MZI arm interference. They can affect output attenuation and, if reflections are appreciable, parasitic interference.

The shortest imbalance, MZI4, provides the largest FSR and the fewest fringes. MZI3 provides the smallest FSR and the most fringe intervals for extracting dispersion. All four satisfy $\text{FSR} < 50 \text{ nm}$ within this model. The minimum imbalance for one 50 nm FSR near 1550 nm is approximately

$$\Delta L_{\min} \approx \frac{(1.55 \mu\text{m})^2}{3.893484(0.050 \mu\text{m})} = 12.34 \mu\text{m}. \quad (1)$$

The existing designs exceed this minimum by factors of about 4.8–13.7. The ratios in Table 1 count approximate fringe peri-

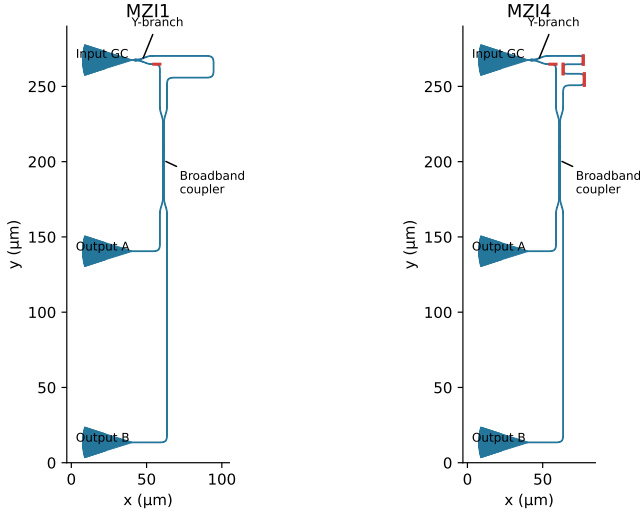


Figure 2: Actual GDS geometry of MZI1 and MZI4. Input light enters the upper grating. The Y-branch splits the light into two arms; the broadband coupler recombines it into two outputs. Red routing markers are retained for review.

Table 1: Actual arm-length variation and approximate performance at 1550 nm; $n_g = 3.893484$. All devices use $w = 500$ nm, $h = 220$ nm, quasi-TE polarization.

Device	L_2 (μ m)	ΔL (μ m)	FSR (nm)	50nm/FSR
MZI1	99.999	68.717	8.980	5.57
MZI2	150.001	118.719	5.198	9.62
MZI3	199.999	168.717	3.657	13.67
MZI4	89.999	58.717	10.509	4.76

ods across the band, not a guaranteed integer count of peaks at arbitrary scan endpoints.

3 Theory

3.1 Propagation phase and group index

For a waveguide mode with propagation constant β ,

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

The second relation follows from $n_g = c d\beta/d\omega$ with $\omega = 2\pi c/\lambda$. For equal waveguide cross-sections in the two arms, the differential phase is

$$\Phi(\lambda) = \beta(\lambda)\Delta L + \phi_c(\lambda), \quad (3)$$

where ϕ_c includes the relative phase introduced by the splitter and combiner. A wavelength-independent ϕ_c shifts fringe positions but does not change the local FSR.

3.2 Transfer function

Let a_1 and a_2 be the positive arm-to-output field-amplitude factors, including input splitting, output combining, and propagation attenuation. A general output transmission is

$$T(\lambda) = a_1^2 + a_2^2 + 2a_1a_2 \cos \Phi(\lambda). \quad (4)$$

For ideal equal splitting and combining, power attenuation coefficient α in reciprocal length units gives $a_j = \frac{1}{2}e^{-\alpha L_j/2}$, hence

$$T_A = \frac{1}{4} [e^{-\alpha L_1} + e^{-\alpha L_2} + 2e^{-\alpha(L_1+L_2)/2} \cos \Phi]. \quad (5)$$

In the lossless case, the two complementary outputs are

$$T_A = \frac{1 + \cos \Phi}{2}, \quad T_B = \frac{1 - \cos \Phi}{2}. \quad (6)$$

These expressions absorb the actual coupler phase into ϕ_c ; a mixed Y-branch/directional-coupler circuit can have a different phase origin from a two-identical-coupler MZI. The calculation below sets $\phi_c = 0$ to illustrate the spectra and test FSR extraction. It does not predict the absolute peak wavelengths of the real circuit.

The fiber-to-fiber transmission is further multiplied by the input and output grating-coupler efficiencies and output-route attenuation. Unequal arm amplitudes reduce fringe visibility,

$$V = \frac{T_{\max} - T_{\min}}{T_{\max} + T_{\min}} = \frac{2a_1a_2}{a_1^2 + a_2^2}. \quad (7)$$

Loss and visibility cannot be reliably forecast from the mask alone, so the presented spectra are normalized lossless predictions, not insertion-loss predictions.

3.3 Derivation of group index from FSR

For adjacent maxima, $|\Delta\Phi| = 2\pi$. Differentiating Eq. 3 and initially treating ϕ_c as constant yields

$$\frac{d\Phi}{d\lambda} = -\frac{2\pi n_g \Delta L}{\lambda^2}. \quad (8)$$

Across a sufficiently narrow fringe interval,

$$\text{FSR}_\lambda \approx \frac{\lambda^2}{n_g |\Delta L|}, \quad \boxed{n_g \approx \frac{\lambda^2}{|\Delta L| \text{FSR}_\lambda}}. \quad (9)$$

All lengths in Eq. 9 must use the same units. In frequency, $d\Phi/df = 2\pi n_g \Delta L/c$, giving

$$\text{FSR}_f \approx \frac{c}{n_g |\Delta L|}. \quad (10)$$

This is the standard unequal-arm MZI relation used in circuit modeling [4].

For adjacent peak wavelengths $\lambda_a < \lambda_b$, use the measured frequency separation instead of approximating it by a center-wavelength conversion:

$$\Delta f = c \left(\frac{1}{\lambda_a} - \frac{1}{\lambda_b} \right), \quad (11)$$

$$\bar{n}_g = \frac{\lambda_a \lambda_b}{|\Delta L|(\lambda_b - \lambda_a)}. \quad (12)$$

With constant coupler phase, Eq. 12 exactly gives the group index averaged over that frequency interval, even if n_g varies across it. It is approximately the local group index at the interval midpoint when dispersion is weak. A peak-to-trough spacing is half an FSR and must not be substituted for a peak-to-peak spacing.

If the combiner phase is dispersive, the extracted quantity is instead an apparent index,

$$n_g^{\text{app}} = n_g + \frac{c}{2\pi\Delta L} \frac{d\phi_c}{df}. \quad (13)$$

More generally the phase slope contains the differential sum of all arm-section group delays, including bends and unequal-width sections. A straight-waveguide index is obtained only when those extra differential contributions cancel or are modeled.

4 Modeling and numerical simulations

4.1 Approximate waveguide model

A two-step effective-index method (EIM) is used as an explicitly approximate calculation. First, the fundamental TE mode of a symmetric 220 nm silicon slab is solved vertically. Its effective index becomes the core index of a lateral 500 nm slab, for which the fundamental TM slab relation is used. This TE-vertical/TM-lateral choice approximates a quasi-TE rectangular guide with the dominant electric field along the width. Both slab exteriors use oxide index 1.444. Silicon index is fixed at 3.476 across the scan. These values are chosen model inputs; material dispersion, sidewall slope, finite oxide thickness, substrate leakage, and corner-vector effects are omitted.

For slab thickness t , indices n_c, n_s , half-thickness $a = t/2$, and fundamental even mode,

$$V_s = \frac{\pi t}{\lambda} \sqrt{n_c^2 - n_s^2}, \quad (14)$$

$$u \tan u = q \sqrt{V_s^2 - u^2}, \quad 0 < u < \min(V_s, \pi/2), \quad (15)$$

$$n_{\text{slab}}^2 = n_c^2 - \left(\frac{u\lambda}{\pi t} \right)^2, \quad (16)$$

where $q = 1$ for TE and $q = n_c^2/n_s^2$ for TM. Root finding on the fundamental branch avoids substituting a higher-order slab mode. The wavelength sweep covers 1525–1575 nm with 501 points; $dn_{\text{eff}}/d\lambda$ is evaluated with a centered $10^{-5} \mu\text{m}$ step. The solver root tolerance is 10^{-14} in u .

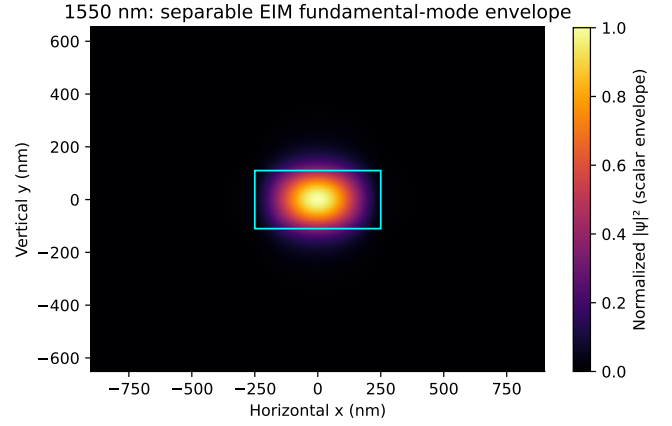


Figure 3: Approximate fundamental-mode scalar envelope at 1550 nm. Cyan outlines the 500 nm × 220 nm silicon core. The plotted quantity is $|\psi|^2$ normalized to its maximum, not a full-vector electric-field intensity or a Lumerical MODE export.

At 1550 nm this model yields

$$n_{\text{eff}} = 2.491433, \quad n_g = 3.893484. \quad (17)$$

These are estimates within the chosen EIM and fixed-material assumptions, not full-vector quasi-TE predictions. In particular, the small computed change in n_g across the band must not be interpreted as a validated low-dispersion result. The compact-fit error and numerical extraction error below test internal numerical consistency, not EIM accuracy.

For the profile in Fig. 3, each slab has a cosine core envelope and an evanescent tail,

$$F(s) = \begin{cases} \cos(us/a), & |s| \leq a, \\ \cos u e^{-v(|s|/a-1)}, & |s| > a, \end{cases} \quad (18)$$

with $v = \sqrt{V_s^2 - u^2}$. The plotted separable approximation is $\psi(x, y) = F_x(x)F_y(y)$. This visualizes confinement, but does not reproduce all vector-field boundary conditions or establish that no higher-order rectangular-guide modes exist.

4.2 Effective/group index and compact model

Figure 4 presents the computed effective and group indices. Define $p = \lambda_{\mu\text{m}} - 1.55$, where $\lambda_{\mu\text{m}}$ is the numerical wavelength in micrometers. A least-squares cubic fit gives

$$n_{\text{eff}}(p) = 2.4914332285 - 0.9045490711p + 0.0012317028p^2 + 0.1498484747p^3. \quad (19)$$

Equivalently, if dimensional p is used, the nonconstant coefficients have units μm^{-1} , μm^{-2} , and μm^{-3} . This fit is intended only for 1.525–1.575 μm and the specified cross-section. The largest effective-index residual over the 501 sample points is 5.34×10^{-9} relative to the EIM data. The fitted group index is obtained analytically from

$$n_g(p) = n_{\text{eff}}(p) - (1.55 + p)(a_1 + 2a_2p + 3a_3p^2), \quad (20)$$

where a_1, a_2, a_3 are the corresponding polynomial coefficients in Eq. 19. No independent group-index fit is required.

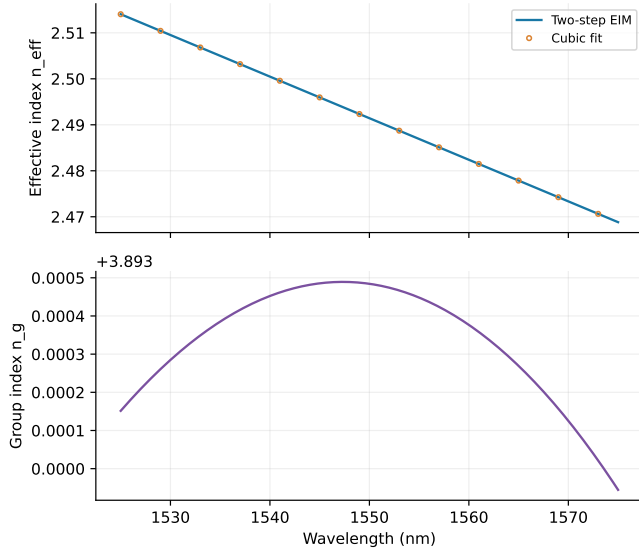


Figure 4: Effective and group index versus wavelength for the approximate oxide-clad EIM model. Open markers compare the cubic fit with the directly solved effective index. Material indices are held fixed; the plotted dispersion is only the model's geometrical waveguide dispersion.

Table 2: Group-index extraction from simulated adjacent maxima. Peak wavelengths are in nm; values shown are rounded from the calculation.

	λ_a	λ_b	Extracted n_g	Midpoint n_g
MZI1	1545.113	1554.088	3.893481	3.893486
MZI2	1549.097	1554.306	3.893474	3.893476
MZI3	1547.049	1550.701	3.893487	3.893488
MZI4	1543.515	1554.007	3.893481	3.893488

4.3 Circuit transmission spectra

The compact model is substituted into Eqs. 3 and 6, using the actual imbalances from Table 1. Spectra are sampled at 20,001 wavelengths across the 50 nm band. Figure 5 displays output A; output B is complementary under the lossless assumptions. The fringe density increases as ΔL increases, while the same waveguide dispersion appears in all devices. The normalized ideal visibility is unity by assumption, not an experimental extinction-ratio prediction.

4.4 Numerical validation of the extraction

Adjacent maxima near 1550 nm were found by solving $\Phi(\lambda) = 2\pi m$ with the directly evaluated EIM index, independently of the sampled spectral grid. Equation 12 was then applied to the peak pairs. Table 2 compares the inferred index with the EIM derivative at each wavelength midpoint. The largest relative difference is below 0.00017%, consistent with interval averaging and weak model dispersion. This demonstrates that the extraction equation and unit conversions recover the simulated group index; it does not validate a real-device model.

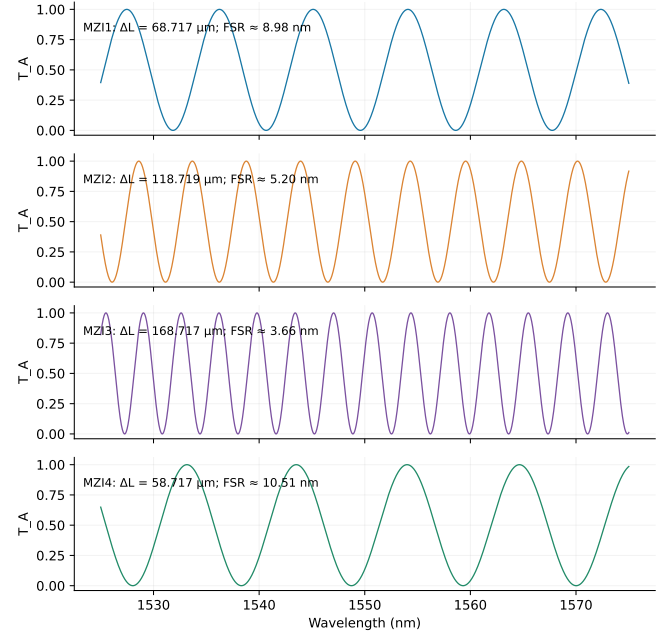


Figure 5: Calculated normalized output-A spectra of all four layout-derived MZIs. These are ideal compact-model simulations with $\alpha = 0$ and $\phi_c = 0$. Coupler/grating S-parameters, reflections, bend losses, and process variations are not included. Actual absolute fringe positions and insertion loss require a calibrated circuit model.

4.5 Full-vector simulation to complete before submission

The course requests mode-profile images and index graphs from MODE and/or MATLAB, and circuit spectra from MATLAB and/or INTERCONNECT. The figures here were generated in Python using the stated approximate model. A MATLAB implementation of the same EIM and circuit calculation is included in the source bundle, but has not been executed in MATLAB. No figure is represented as a MODE, MATLAB, or INTERCONNECT export.

For a submission requiring those specific tools, run the supplied MATLAB implementation to produce the approximate-model figures, or preferably replace them with a full-vector MODE calculation. In MODE, construct the 500 nm $\times$ 220 nm silicon core with the actual foundry cladding and dispersive material data; solve for the fundamental quasi-TE mode; record vector fields and TE fraction; and track the same mode across 1525–1575 nm. Refine the mesh and enlarge the domain until n_{eff} and n_g converge to the chosen tolerance (a proposed target is 10^{-3} in n_g). Refit Eq. 19 and regenerate Table 1 using those data. For INTERCONNECT, use the mixed Y-branch/broadband-coupler topology and component S-parameters, not two generic Y-splitters. Ansys documents mode tracking, dispersion export, and imported component models in its circuit workflow [4].

5 Fabrication and layout review

5.1 Proposed fabrication approach

The design targets a passive 220 nm silicon-on-insulator process with electron-beam-defined silicon patterning and oxide passivation, consistent with the published OpenEBL process overview [3]. The course/foundry run defines the actual resist, etch chemistry, etch depth, oxide thickness, and wafer stack; these details cannot be inferred from the mask. Silicon waveguide polygons occur on layer 1/0, automated-test labels on layer 10, and the floorplan on layer 99. Other drawing/metadata layers must be interpreted according to the matching EBeam technology file rather than treated as extra fabrication masks.

The present report is a design proposal. Fabrication dates, as-built widths, etch depth, sidewall measurements, and microscopy will be added after processing. Width and thickness changes perturb n_{eff} and its wavelength derivative. Roughness and tight bends can increase loss, alter amplitude balance, and lower fringe contrast. Both effects should be considered when comparing a straight-waveguide simulation with a fabricated interferometer.

5.2 Routing issues requiring review

The GDS contains layer-999/0 paths in the common short arm and the MZI4 long arm. In the short arm, the first waypoint segment is only $4.090\ \mu\text{m}$ while the stated radius parameter is $5\ \mu\text{m}$. For MZI4, the internal segments between successive right-angle turns are 5.550 , 6.310 , and $7.640\ \mu\text{m}$, each shorter than the $2R = 10\ \mu\text{m}$ tangent distance needed for two ordinary $5\ \mu\text{m}$ circular fillets. These observations are geometric checks, not a claim about the exact proprietary/PDK bend-construction algorithm. They motivate inspecting the regenerated PCells and marker messages with the original technology version.

The layout also stores a historical “0 errors” SiEPIC-Tools verification label dated 4 October 2026 (SiEPIC-Tools v0.5.31, EBeam v0.4.53, KLayout v0.28.15). This is a recorded verification result, not a fresh signoff performed for this report. Before fabrication, review the layer-999 markers, confirm allowed bend geometry, rerun the matching PDK checks, and re-extract path lengths after any routing change. A corrected route may change the FSR predictions.

The four current optical measurement labels use the course-style word device. The current public OpenEBL instructions instead show openEBL in that position [3]. The correct convention should be checked against the specific course submission rules; no automatic rename is assumed here. The $605\ \mu\text{m} \times 410\ \mu\text{m}$ floorplan alone does not prove the design is ready for a particular foundry/test submission.

6 Experimental plan and later analysis

6.1 Measurement procedure

Use a tunable source over an approximately 50 nm band around 1550 nm and adjust the input polarization to excite the TE grating coupler. Align the input fiber to the labeled upper grating and record the transmitted power at both output gratings when the test system permits. Keep temperature and polarization stable. A proposed wavelength step of 0.02 nm provides over 180 samples per predicted shortest FSR; verify the instrument’s actual resolution and calibration before using this setting.

Measure a grating-to-grating reference if one is available on the shared chip. The uploaded design does not contain such a reference. Normalize the MZI spectrum by this reference, or estimate a slowly varying grating envelope when no reference exists. Avoid treating normalization as a substitute for a reflection model if parasitic cavities are present. Because the two output access routes differ in length, their measured powers need not sum to a wavelength-independent constant without separate output calibration.

Find consecutive maxima (or consecutive minima) of the same fringe family. Fit each peak locally to estimate its center rather than selecting only the largest sampled point. Convert each adjacent pair to Δf , apply Eq. 12, and assign the interval-average index to its wavelength midpoint. Report $n_g(\lambda)$ for all usable intervals and compare the two outputs where possible. Peak and trough families should agree within measurement uncertainty; disagreement can reveal envelope distortion, reflections, or poor fits.

6.2 Cross-device analysis and uncertainty

At a common wavelength, compare the measured FSRs with the inverse- ΔL prediction. Rather than averaging unlike intervals without accounting for dispersion, interpolate the extracted n_g values to common wavelengths. A useful additional analysis fits differential delay against imbalance:

$$\frac{1}{\text{FSR}_f} = \frac{n_g}{c} \Delta L + \tau_c, \quad (21)$$

where $\tau_c = (2\pi)^{-1} d\phi_c/df$ represents a common differential component delay if shared component models and matched bend contributions justify that assumption. The slope yields n_g/c ; a nonzero intercept motivates reviewing coupler phase and route contributions. The MZI4 bend pattern differs from the other long arms, so this fit must be interpreted with care.

For independent uncertainties in the frequency interval and imbalance,

$$\left(\frac{\sigma_{n_g}}{n_g} \right)^2 \approx \left(\frac{\sigma_{\Delta f}}{\Delta f} \right)^2 + \left(\frac{\sigma_{\Delta L}}{\Delta L} \right)^2. \quad (22)$$

If peak-center errors are independent with equal uncertainty σ_λ , then $\sigma_{\text{FSR}_\lambda} \approx \sqrt{2}\sigma_\lambda$. Shared wavelength calibration errors, temperature drift, and coupler/bend model errors may

be correlated or systematic and should be accounted for separately. Numerical root-finding precision is not an experimental uncertainty estimate.

6.3 Sections to add after fabrication

Experimental data: include raw and normalized spectra, scan settings, polarization, temperature, calibration, and device identifiers. **Analysis:** report per-interval FSR and n_g , uncertainties, visibility/extinction, and cross-device comparisons with updated full-vector simulations. **Discussion:** assess process-induced dimension changes, bend contributions, component phase, parasitic reflections, and model limitations. These sections remain prospective; no measurement or fabrication outcome is asserted.

7 Conclusion

The supplied four-MZI layout provides a practical path-length sweep for extracting the group index of a 500 nm-wide quasi-TE silicon strip waveguide with the course-specified 220 nm thickness. The correctly calculated imbalances yield approximate FSRs of 3.66–10.51 nm, comfortably below the available measurement bandwidth. The numerical test recovers the approximate model's group index from adjacent simulated maxima. Final submission should confirm the public username, complete the required simulation-tool outputs, review routing markers, and update lengths if the mask changes. Experimental results will then test the modeled group index and the expected FSR scaling.

Reproducibility and source provenance

The accompanying source bundle contains the LaTeX report, generated figures, a Python EIM/circuit calculation, a MATLAB EIM implementation, extracted parameter/index/spectrum CSV files, and the original GDS. The original mask was read without modification. All layout pictures are derived from its silicon polygons. All simulated plots are original calculations made for this proposal. The design requirements and 50 nm measurement bandwidth are taken from the course text supplied with the request. References below support the relevant component lineage, process context, and circuit equations; they do not certify the current layout or the approximate simulation values.

References

- [1] W. Huynh, *EBeam_HuynhWSiPho.gds*, supplied GDSII layout, top cell EBeam_HuynhWSiPho_v1; inspected 6 Oct. 2026.
- [2] edX Phot1x, “Design Proposal–Report,” course instructions supplied with the design request, accessed 6 Oct. 2026.
- [3] SiEPIC, “OpenEBL,” fabrication process and submission instructions. <https://siepic.ca/openEBL/>, accessed 6 Oct. 2026.
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- [6] Y. Wang *et al.*, “Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits,” *Opt. Express*, vol. 22, pp. 20652–20662, 2014. doi:10.1364/OE.22.020652. DOI and design attribution are embedded in the grating-coupler cell.